Vector Spaces and Subspaces

Definitions, Examples, Linear Combinations and Span

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1 Vector spaces

Throughout, $\mathbb{F}$ denotes the field of scalars. For concreteness the reader may take $\mathbb{F} = \mathbb{R}$ (real numbers); every statement below holds verbatim for $\mathbb{F} = \mathbb{C}$ or any other field.

Definition 1.1 (Vector space)**.** A *vector space* over $\mathbb{F}$ is a nonempty set V, whose elements are called *vectors*, together with two operations

$$\text{addition } V \times V \to V,\ (\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u} + \mathbf{v}, \qquad \text{scalar multiplication } \mathbb{F} \times V \to V,\ (c, \mathbf{v}) \mapsto c\mathbf{v},$$

such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{F}$:

(A1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity)

(A2) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity)

(A3) there is a vector $\mathbf{0} \in V$ with $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v}$ (zero vector)

(A4) for each $\mathbf{v}$ there is $-\mathbf{v} \in V$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ (additive inverse)

(M1) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ (distributivity over vectors)

(M2) $(a+b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ (distributivity over scalars)

(M3) $a(b\mathbf{v}) = (ab)\mathbf{v}$ (compatibility)

(M4) $1\mathbf{v} = \mathbf{v}$ (identity)

Remark 1.2. The phrase "operations $V \times V \to V$ and $\mathbb{F} \times V \to V$" already contains the two *closure* requirements: the sum of two vectors and a scalar multiple of a vector must again lie in V. Many textbooks list these separately as axioms (C1)–(C2), giving "ten axioms". When checking whether a given set is a vector space, closure is usually the first thing to verify.

How to read the definition

A vector space is *any* collection of objects that can be added and scaled so that the familiar rules of arithmetic hold. The objects need not be arrows or lists of numbers: matrices, polynomials and functions are vectors too. What matters is the operations, not the nature of the objects.

1.1 Basic consequences of the axioms

Proposition 1.3. *Let V be a vector space, $\mathbf{v} \in V$ and $c \in \mathbb{F}$. Then*

(a) the zero vector is unique;

(b) the additive inverse of $\mathbf{v}$ is unique;

(c) $0\mathbf{v} = \mathbf{0}$;

(d) $c\mathbf{0} = \mathbf{0}$;

(e) $(-1)\mathbf{v} = -\mathbf{v}$;

(f) if $c\mathbf{v} = \mathbf{0}$, then $c = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. (a) If $\mathbf{0}$ and $\mathbf{0}'$ both satisfy (A3), then $\mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}$, using (A3) for $\mathbf{0}$, then (A1), then (A3) for $\mathbf{0}'$.

(b) If $\mathbf{w}$ and $\mathbf{w}'$ are both inverses of $\mathbf{v}$, then $\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}'$.

(c) By (M2), $0\mathbf{v} = (0+0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$. Adding $-(0\mathbf{v})$ to both sides and using (A2), (A4), (A3) gives $\mathbf{0} = 0\mathbf{v}$.

(d) By (A3) and (M1), $c\mathbf{0} = c(\mathbf{0} + \mathbf{0}) = c\mathbf{0} + c\mathbf{0}$; cancel as in (c).

(e) $\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}$ by (M4), (M2) and (c). By uniqueness (b), $(-1)\mathbf{v} = -\mathbf{v}$.

(f) Suppose $c\mathbf{v} = \mathbf{0}$ and $c \neq 0$. Then $\mathbf{v} = 1\mathbf{v} = (c^{-1}c)\mathbf{v} = c^{-1}(c\mathbf{v}) = c^{-1}\mathbf{0} = \mathbf{0}$ by (M4), (M3) and (d). □

1.2 Examples of vector spaces

Example 1.4 (The coordinate space $\mathbb{R}^n$)**.** $\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ with componentwise operations

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \qquad c(x_1, \dots, x_n) = (cx_1, \dots, cx_n).$$

The zero vector is $(0, \dots, 0)$ and $-(x_1, \dots, x_n) = (-x_1, \dots, -x_n)$. Each axiom reduces to the corresponding rule for real numbers in each component. For $n = 2, 3$ these are the familiar plane and space of arrows.

Example 1.5 (Matrices)**.** The set $M_{m\times n}(\mathbb{R})$ of all $m \times n$ real matrices, with entrywise addition and scalar multiplication, is a vector space. Its zero vector is the zero matrix. For instance in $M_{2\times 2}(\mathbb{R})$,

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 3 \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 0 & 10 \end{pmatrix}.$$

Example 1.6 (Polynomials)**.** P_n denotes the set of real polynomials of degree at most n,

$$P_n = \{a_0 + a_1 x + \cdots + a_n x^n : a_0, \ldots, a_n \in \mathbb{R}\},$$

with the usual addition and multiplication by constants. The sum of two polynomials of degree $\leq n$ has degree $\leq n$, so closure holds. The zero vector is the zero polynomial. The set P of *all* polynomials is also a vector space.

Example 1.7 (Function spaces)**.** Let X be any nonempty set and let $\mathcal{F}(X)$ be the set of all functions $f : X \to \mathbb{R}$, with pointwise operations

$$(f + g)(x) = f(x) + g(x), \qquad (cf)(x) = c\, f(x).$$

The zero vector is the function identically equal to 0. Taking $X = \{1, \ldots, n\}$ recovers $\mathbb{R}^n$; taking $X = \{1, 2, 3, \ldots\}$ gives the space of real sequences. The set $C[a, b]$ of continuous functions on $[a, b]$ is a vector space because sums and multiples of continuous functions are continuous.

Example 1.8 (An unusual vector space)**.** Let $V = (0, \infty)$, the positive reals, with

$$x \oplus y = xy, \qquad c \odot x = x^c.$$

This *is* a vector space over $\mathbb{R}$. The "zero vector" is the number 1 (because $x \oplus 1 = x$), and the "negative" of x is $1/x$. For example, (M2) reads $x^{a+b} = x^a x^b$ and (M1) reads $(xy)^a = x^a y^a$. The example shows that the zero vector need not be the number 0.

Non-example 1.9 (A failed axiom)**.** On $\mathbb{R}^2$ keep the usual addition but define $c(x, y) = (cx, 0)$. Axioms (A1)–(A4), (M1), (M3) hold, and so does (M2): $(a + b)(x, y) = ((a + b)x, 0) = (ax, 0) + (bx, 0)$. But (M4) fails: $1(0, 1) = (0, 0) \neq (0, 1)$. Hence this is *not* a vector space. One failed axiom is enough.

Non-example 1.10. The set $\mathbb{Z}^2$ of integer pairs is not a vector space over $\mathbb{R}$: $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin \mathbb{Z}^2$, so scalar multiplication is not closed.

1.3 The zero space

Definition 1.11 (Zero space)**.** The set $\{\mathbf{0}\}$ consisting of a single element, with $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $c\mathbf{0} = \mathbf{0}$ for every $c \in \mathbb{F}$, is a vector space called the *zero space* (or *trivial vector space*).

All axioms hold trivially because every expression evaluates to $\mathbf{0}$. It is the smallest possible vector space: by (A3), every vector space contains at least the zero vector, so a vector space is never empty. Note the distinction between the zero *vector* $\mathbf{0}$, the zero *scalar* 0, and the zero *space* $\{\mathbf{0}\}$. The empty set $\emptyset$ is **not** a vector space, since it has no zero vector.

2 Subspaces

Definition 2.1 (Subspace)**.** A subset W of a vector space V is a *subspace* of V if W is itself a vector space under the addition and scalar multiplication inherited from V.

Checking all eight axioms is unnecessary: those that are identities ((A1), (A2), (M1)–(M4)) are automatically inherited from V. Only the closure properties and the presence of $\mathbf{0}$ need checking.

Theorem 2.2 (Subspace test)**.** *A subset $W \subseteq V$ is a subspace of V if and only if*

(S1) $\mathbf{0} \in W$*;*

(S2) $\mathbf{u}, \mathbf{v} \in W \implies \mathbf{u} + \mathbf{v} \in W$ *(closed under addition);*

(S3) $c \in \mathbb{F},\ \mathbf{v} \in W \implies c\mathbf{v} \in W$ *(closed under scalar multiplication).*

Equivalently: $W \neq \emptyset$ *and* $a\mathbf{u} + b\mathbf{v} \in W$ *for all* $\mathbf{u}, \mathbf{v} \in W$*,* $a, b \in \mathbb{F}$*.*

Proof. If W is a subspace, it contains a zero vector $\mathbf{0}_W$; then $\mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$ in V, and adding $-\mathbf{0}_W$ (in V) gives $\mathbf{0}_W = \mathbf{0}$. So (S1) holds, and (S2), (S3) are the closure requirements.

Conversely, assume (S1)–(S3). Then addition and scalar multiplication map into W. Axioms (A1), (A2), (M1)–(M4) hold in W because they hold in V. (A3) holds because $\mathbf{0} \in W$. For (A4), if $\mathbf{v} \in W$ then $-\mathbf{v} = (-1)\mathbf{v} \in W$ by (S3) and Proposition 1.3(e).

For the equivalent form: (S1)–(S3) imply $a\mathbf{u} + b\mathbf{v} \in W$. Conversely, if $W \neq \emptyset$ and W is closed under such combinations, pick $\mathbf{w} \in W$; then $\mathbf{0} = 0\mathbf{w} + 0\mathbf{w} \in W$, and taking $b = 0$ or $a = b = 1$ gives (S3) and (S2). □

Quick rejection

If $\mathbf{0} \notin W$, then W is not a subspace. This is often the fastest check: for example, any line or plane in $\mathbb{R}^3$ that does not pass through the origin is not a subspace.

2.1 Examples of subspaces

Example 2.3 (Trivial subspaces)**.** Every vector space V has the two subspaces $\{\mathbf{0}\}$ (the zero subspace) and V itself. Any other subspace is called *proper and nontrivial*.

Example 2.4 (A line through the origin)**.** $W = \{(x, y) \in \mathbb{R}^2 : y = 2x\}$. Here $(0, 0) \in W$. If $(x_1, 2x_1), (x_2, 2x_2) \in W$, their sum is $(x_1 + x_2,\ 2(x_1 + x_2)) \in W$, and $c(x, 2x) = (cx, 2(cx)) \in W$. So W is a subspace of $\mathbb{R}^2$.

Example 2.5 (A plane through the origin)**.** $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$. The zero vector satisfies the equation. If $\mathbf{u} = (x_1, y_1, z_1)$ and $\mathbf{v} = (x_2, y_2, z_2)$ satisfy it, then

$$(x_1 + x_2) - 2(y_1 + y_2) + (z_1 + z_2) = (x_1 - 2y_1 + z_1) + (x_2 - 2y_2 + z_2) = 0 + 0 = 0,$$

and $cx_1 - 2cy_1 + cz_1 = c \cdot 0 = 0$. So W is a subspace of $\mathbb{R}^3$. (We will meet this plane again in Example 4.4.)

Example 2.6 (Null space of a matrix)**.** Let A be an $m \times n$ matrix. The solution set of the homogeneous system,

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\},$$

is a subspace of $\mathbb{R}^n$: $A\mathbf{0} = \mathbf{0}$; if $A\mathbf{x} = A\mathbf{y} = \mathbf{0}$ then $A(\mathbf{x} + \mathbf{y}) = \mathbf{0}$ and $A(c\mathbf{x}) = cA\mathbf{x} = \mathbf{0}$. The previous two examples are special cases.

Example 2.7 (Subspaces of matrices)**.** In $M_{n\times n}(\mathbb{R})$, the symmetric matrices ($A^T = A$), the diagonal matrices, the upper triangular matrices, and the matrices of trace zero are all subspaces. For the last: $\operatorname{tr}(0) = 0$, $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$ and $\operatorname{tr}(cA) = c \operatorname{tr} A$.

Example 2.8 (Subspaces of function spaces)**.** P_n is a subspace of P_{n+1}, and of P. Inside $C[0, 1]$, the set $\{f \in C[0, 1] : f(0) = 0\}$ is a subspace; so is the set of differentiable functions, and the set of solutions of the differential equation $f'' + f = 0$.

Non-example 2.9 (Three ways to fail)**.**

(i) $W_1 = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$: no zero vector, since $0 + 0 \neq 1$.

(ii) $W_2 = \{(x, y) \in \mathbb{R}^2 : x \geq 0\}$ (a half-plane): contains $\mathbf{0}$ and is closed under addition, but $(-1)(1, 0) = (-1, 0) \notin W_2$.

(iii) $W_3 = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ (the union of the two axes): contains $\mathbf{0}$ and is closed under scalar multiplication, but $(1, 0) + (0, 1) = (1, 1) \notin W_3$.

Similarly, the solution set of a *non*homogeneous system $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is never a subspace (it misses $\mathbf{0}$), and $\{p \in P_2 : \deg p = 2\}$ is not a subspace (it misses the zero polynomial, and $x^2 + (-x^2) = 0$).

Remark 2.10 (The subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$). The subspaces of $\mathbb{R}^2$ are exactly: $\{\mathbf{0}\}$, lines through the origin, and $\mathbb{R}^2$. The subspaces of $\mathbb{R}^3$ are exactly: $\{\mathbf{0}\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$. (The proof uses the notion of dimension, which comes after span and linear independence.)

2.2 Intersections and unions

Proposition 2.11. *If W_1 and W_2 are subspaces of V, then $W_1 \cap W_2$ is a subspace of V. More generally, any intersection of subspaces is a subspace.*

Proof. $\mathbf{0}$ lies in every subspace, hence in the intersection. If $\mathbf{u}, \mathbf{v}$ lie in every subspace of the family, so do $\mathbf{u} + \mathbf{v}$ and $c\mathbf{v}$, by (S2) and (S3) applied in each one. □

Remark 2.12. A union of two subspaces is in general *not* a subspace: W_3 in the non-example above is the union of the x-axis and the y-axis. In fact $W_1 \cup W_2$ is a subspace only when $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$ (Exercise 6.6).

3 Linear combinations

Definition 3.1 (Linear combination)**.** Let $\mathbf{v}_1, \ldots, \mathbf{v}_k \in V$. A *linear combination* of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ is any vector of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k, \qquad c_1, \ldots, c_k \in \mathbb{F}.$$

The scalars c_i are called the *coefficients* (or *weights*) of the combination.

The zero vector is a linear combination of any vectors (take all $c_i = 0$), and each $\mathbf{v}_i$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ (take $c_i = 1$, all others 0).

Example 3.2 (Computing a linear combination)**.** In $\mathbb{R}^3$ with $\mathbf{v}_1 = (1, 2, 3)$ and $\mathbf{v}_2 = (2, 1, 0)$,

$$3\mathbf{v}_1 - 2\mathbf{v}_2 = (3, 6, 9) - (4, 2, 0) = (-1, 4, 9).$$

The central computational question is the reverse one:

Is $\mathbf{b}$ a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$?

In $\mathbb{R}^n$, this asks whether the vector equation $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{b}$ has a solution. Writing the $\mathbf{v}_i$ as the columns of a matrix $A = [\,\mathbf{v}_1 \ \cdots \ \mathbf{v}_k\,]$, it is the linear system $A\mathbf{c} = \mathbf{b}$. So: $\mathbf{b}$ *is a linear combination of the columns of A if and only if the system $A\mathbf{c} = \mathbf{b}$ is consistent.*

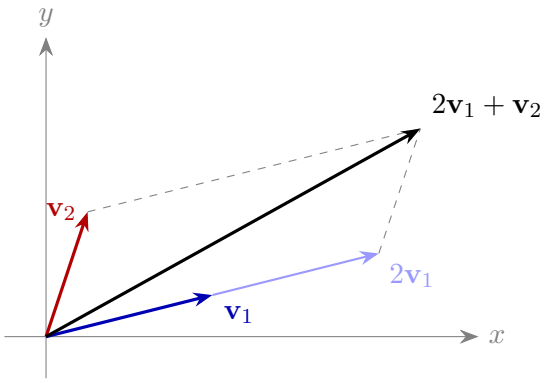


Figure 1: The linear combination $2\mathbf{v}_1 + \mathbf{v}_2$: walk twice along $\mathbf{v}_1$, then once along $\mathbf{v}_2$ (parallelogram rule).

Example 3.3 (Yes)**.** Is $\mathbf{b} = (-1, 4, 9)$ a linear combination of $\mathbf{v}_1 = (1, 2, 3)$ and $\mathbf{v}_2 = (2, 1, 0)$? We solve

$$c_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix} \iff \begin{cases} c_1 + 2c_2 = -1, \\ 2c_1 + c_2 = 4, \\ 3c_1 = 9. \end{cases}$$

The third equation gives $c_1 = 3$; the first then gives $c_2 = -2$; and the second checks: $2(3) + (-2) = 4$. So $\mathbf{b} = 3\mathbf{v}_1 - 2\mathbf{v}_2$.

Example 3.4 (No)**.** Is $\mathbf{b} = (1, 0, 0)$ a linear combination of the same $\mathbf{v}_1, \mathbf{v}_2$? The system is $c_1 + 2c_2 = 1$, $2c_1 + c_2 = 0$, $3c_1 = 0$. The third equation gives $c_1 = 0$, the second then gives $c_2 = 0$, and the first becomes $0 = 1$, a contradiction. So $(1, 0, 0)$ is *not* a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Example 3.5 (Polynomials)**.** In P_2, is $p(x) = 1$ a linear combination of $p_1 = 1 + x$, $p_2 = x + x^2$, $p_3 = 1 + x^2$? Compare coefficients in

$$a(1 + x) + b(x + x^2) + c(1 + x^2) = (a + c) + (a + b)x + (b + c)x^2 = 1.$$

This requires $a + c = 1$, $a + b = 0$, $b + c = 0$, so $b = -a$, $c = a$, $2a = 1$. Hence

$$1 = \tfrac{1}{2}(1 + x) - \tfrac{1}{2}(x + x^2) + \tfrac{1}{2}(1 + x^2).$$

Example 3.6 (Matrices)**.** In $M_{2\times 2}(\mathbb{R})$,

$$\begin{pmatrix} 3 & -1 \\ -1 & 5 \end{pmatrix} = 3 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 5 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} - 1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Example 3.7 (Functions)**.** In $\mathcal{F}(\mathbb{R})$, the identities $\cos^2 x + \sin^2 x = 1$ and $\cos^2 x - \sin^2 x = \cos 2x$ say that the constant function 1 and the function $\cos 2x$ are both linear combinations of $\cos^2 x$ and $\sin^2 x$.

4 Span

Definition 4.1 (Span)**.** Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be a finite subset of V. The *span* of S is the set of all linear combinations of its elements:

$$\operatorname{span} S = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_1, \ldots, c_k \in \mathbb{F}\}.$$

For an arbitrary (possibly infinite) subset $S \subseteq V$, $\operatorname{span} S$ is the set of all linear combinations of *finitely many* vectors of S. By convention, $\operatorname{span} \emptyset = \{\mathbf{0}\}$.

If $\operatorname{span} S = W$, we say that S *spans* (or *generates*) W, and that S is a *spanning set* for W.

So "$\mathbf{b} \in \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$" is just another way of saying "$\mathbf{b}$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$". Examples 3.3 and 3.4 show that

$$(-1, 4, 9) \in \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\} \quad \text{and} \quad (1, 0, 0) \notin \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}.$$

4.1 The span is a subspace

Theorem 4.2. *Let $S \subseteq V$. Then*

(a) $\operatorname{span} S$ *is a subspace of V containing S;*

(b) if W is any subspace of V with $S \subseteq W$, then $\operatorname{span} S \subseteq W$.

Hence $\operatorname{span} S$ *is the* smallest *subspace of V that contains S; it equals the intersection of all subspaces containing S.*

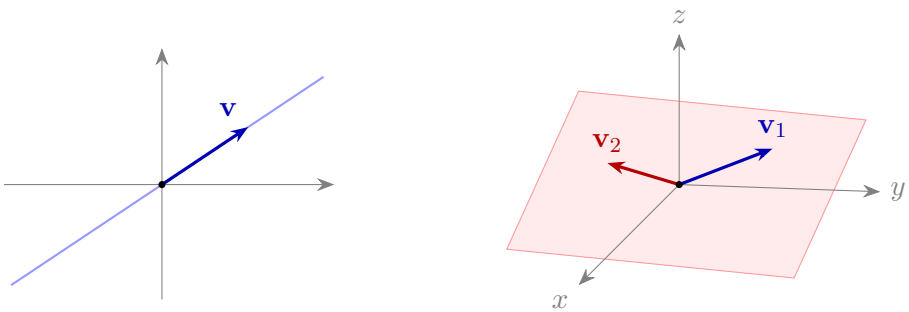
Proof. (a) If $S = \emptyset$ the claim is $\{\mathbf{0}\}$, a subspace. Otherwise: $\mathbf{0} = 0\mathbf{v}$ for any $\mathbf{v} \in S$, so $\mathbf{0} \in \operatorname{span} S$. A sum of two linear combinations of finitely many vectors of S is again such a combination (collect terms involving the same vector), and $c\,(c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = (cc_1)\mathbf{v}_1 + \cdots + (cc_k)\mathbf{v}_k$. By Theorem 2.2, $\operatorname{span} S$ is a subspace. Each $\mathbf{v} \in S$ equals $1\mathbf{v}$, so $S \subseteq \operatorname{span} S$.

(b) A subspace $W \supseteq S$ is closed under addition and scalar multiplication, so by induction on the number of terms it contains every $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ with $\mathbf{v}_i \in S$.

For the last statement: the intersection of all subspaces containing S is a subspace containing S, hence contains $\operatorname{span} S$ by (b); and $\operatorname{span} S$ is one of the subspaces being intersected, by (a), so the intersection is contained in $\operatorname{span} S$. □

Geometric meaning in $\mathbb{R}^2$ and $\mathbb{R}^3$

- $\operatorname{span}\{\mathbf{0}\} = \{\mathbf{0}\}$, a point.
- $\operatorname{span}\{\mathbf{v}\}$ with $\mathbf{v} \neq \mathbf{0}$ is the line through the origin in the direction of $\mathbf{v}$: all multiples $t\mathbf{v}$.
- $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, where neither vector is a multiple of the other, is the plane through the origin containing both vectors. If one is a multiple of the other, the span is only a line.
- In $\mathbb{R}^3$, three vectors that do not lie in a common plane through the origin span all of $\mathbb{R}^3$.



$\operatorname{span}\{\mathbf{v}\}$: a line through $\mathbf{0}$ $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$: a plane through $\mathbf{0}$

Figure 2: Spans in low dimensions.

4.2 Examples of span

Example 4.3 (Standard vectors)**.** In $\mathbb{R}^3$ let $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$, $\mathbf{e}_3 = (0, 0, 1)$. Every (x, y, z) equals $x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$, so $\operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3$. Also $\operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \{(x, y, 0)\}$ is the xy-plane, and $\operatorname{span}\{\mathbf{e}_1\}$ is the x-axis.

Example 4.4 (Describing a span by an equation)**.** Describe $W = \operatorname{span}\{(1,2,3),(2,1,0)\}$ in $\mathbb{R}^3$. A vector (x,y,z) lies in W exactly when the system

$$c_1 + 2c_2 = x, \qquad 2c_1 + c_2 = y, \qquad 3c_1 = z$$

is consistent. From the third equation $c_1 = z/3$; from the second $c_2 = y - 2z/3$. Substituting in the first:

$$\tfrac{z}{3} + 2y - \tfrac{4z}{3} = x \iff x - 2y + z = 0.$$

Therefore $W = \{(x,y,z) : x - 2y + z = 0\}$, a plane through the origin. This explains Examples 3.3–3.4: $(-1,4,9)$ satisfies $-1 - 8 + 9 = 0$, while $(1,0,0)$ gives $1 \neq 0$.

Example 4.5 (A spanning set for $\mathbb{R}^2$)**.** Do $\mathbf{u} = (1,1)$ and $\mathbf{v} = (1,-1)$ span $\mathbb{R}^2$? Given any (a,b), solve $c_1 + c_2 = a$, $c_1 - c_2 = b$: $c_1 = \frac{a+b}{2}$, $c_2 = \frac{a-b}{2}$. A solution always exists, so $\operatorname{span}\{\mathbf{u},\mathbf{v}\} = \mathbb{R}^2$. By contrast, $\operatorname{span}\{(1,2),(2,4)\}$ is only the line $y = 2x$, since $(2,4) = 2(1,2)$.

Example 4.6 (A spanning set for P_2)**.** The monomials $1, x, x^2$ span P_2 by definition. Less obviously, $\{1+x,\ x+x^2,\ 1+x^2\}$ also spans P_2: as in Example 3.5, for any $q = \alpha + \beta x + \gamma x^2$ the system $a + c = \alpha$, $a + b = \beta$, $b + c = \gamma$ has the solution

$$a = \tfrac{\alpha+\beta-\gamma}{2}, \qquad b = \tfrac{-\alpha+\beta+\gamma}{2}, \qquad c = \tfrac{\alpha-\beta+\gamma}{2}.$$

Example 4.7 (Spans of matrices)**.** In $M_{2\times 2}(\mathbb{R})$, write $E_{11}, E_{12}, E_{21}, E_{22}$ for the matrices with a single entry 1 in the indicated position and 0 elsewhere. Then

- $\operatorname{span}\{E_{11}, E_{12}, E_{21}, E_{22}\} = M_{2\times 2}(\mathbb{R})$;
- $\operatorname{span}\{E_{11}, E_{22}\}$ is the subspace of diagonal matrices;
- $\operatorname{span}\{E_{11}, E_{22}, E_{12}+E_{21}\}$ is the subspace of symmetric matrices, because $\begin{pmatrix} a & b \\ b & d \end{pmatrix} = aE_{11} + dE_{22} + b(E_{12} + E_{21})$.

Example 4.8 (An infinite spanning set)**.** The set $\{1, x, x^2, x^3, \dots\}$ spans the space P of all polynomials: every polynomial is a *finite* linear combination of monomials. No finite set spans P, because finitely many polynomials have a maximal degree d, and x^{d+1} is not in their span.

4.3 When does adding a vector enlarge the span?

Proposition 4.9. *Let* $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w} \in V$. *Then*

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\} = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \iff \mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}.$$

Proof. Write $U = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ and $U' = \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\}$. Always $U \subseteq U'$ (take coefficient 0 on $\mathbf{w}$), and $\mathbf{w} \in U'$. ($\Rightarrow$) If $U' = U$, then $\mathbf{w} \in U$. ($\Leftarrow$) If $\mathbf{w} \in U$, then U is a subspace containing $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}$, so $U' \subseteq U$ by Theorem 4.2(b). □

So a vector that is already a linear combination of the others is *redundant* for the span. For example $\operatorname{span}\{(1,2),(2,4)\} = \operatorname{span}\{(1,2)\}$, and

$$\operatorname{span}\{(1,2,3),(2,1,0),(-1,4,9)\} = \operatorname{span}\{(1,2,3),(2,1,0)\}.$$

Removing redundant vectors until none is left leads to the ideas of *linear independence* and *basis*, the next topics.

Corollary 4.10. *The span does not change if one (i) swaps two vectors, (ii) multiplies a vector by a nonzero scalar, or (iii) adds a multiple of one vector to another. For instance* $\operatorname{span}\{\mathbf{u},\mathbf{v}\} = \operatorname{span}\{\mathbf{u}, \mathbf{v} + c\mathbf{u}\}$.

Proof. In each case every new vector is a linear combination of the old ones, and vice versa (the operations can be undone by operations of the same type). By Theorem 4.2(b), each span contains the other. □

5 Summary

- A **vector space** is a set with addition and scalar multiplication satisfying the axioms (A1)–(A4), (M1)–(M4).
- The **zero space** $\{\mathbf{0}\}$ is the smallest vector space; every vector space contains $\mathbf{0}$.
- A **subspace** is a subset containing $\mathbf{0}$ and closed under addition and scalar multiplication.
- A **linear combination** is $c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k$. Deciding whether $\mathbf{b}$ is one amounts to solving a linear system.
- The **span** of S is the set of all linear combinations of vectors in S; it is the smallest subspace containing S.

6 Exercises

Exercise 6.1. Decide whether each set is a subspace of $\mathbb{R}^3$. Justify your answer.

(a) $\{(x, y, z) : x = 3z\}$

(b) $\{(x, y, z) : x + y + z = 2\}$

(c) $\{(x, y, z) : x^2 = y^2\}$

(d) $\{(t, 2t, -t) : t \in \mathbb{R}\}$

Exercise 6.2. Is $\mathbf{b} = (4, 3, 5)$ in $\operatorname{span}\{(1, 0, 1), (2, 1, 3)\}$? If so, write it as a linear combination.

Exercise 6.3. Find an equation describing $\operatorname{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Exercise 6.4. Show that $\{1,\ 1 + x,\ 1 + x + x^2\}$ spans P_2.

Exercise 6.5. Is the set of invertible 2×2 matrices a subspace of $M_{2\times2}(\mathbb{R})$?

Exercise 6.6. Let W_1, W_2 be subspaces of V. Prove that $W_1 \cup W_2$ is a subspace if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Solutions

1. (a) Yes: it is the null space of $(1\ \ 0\ \ -3)$. (b) No: $\mathbf{0}$ does not satisfy $0 = 2$. (c) No: $(1, 1, 0)$ and $(1, -1, 0)$ belong, but their sum $(2, 0, 0)$ does not. (d) Yes: it is $\operatorname{span}\{(1, 2, -1)\}$, a line through the origin.

2. Solve $c_1 + 2c_2 = 4$, $c_2 = 3$, $c_1 + 3c_2 = 5$. The second gives $c_2 = 3$, the first $c_1 = -2$, and the third checks: $-2 + 9 = 7 \neq 5$. So $\mathbf{b} \notin$ span. (By contrast, $(4, 3, 7) = -2(1, 0, 1) + 3(2, 1, 3)$ is.)

3. $(x, y, z) = c_1(1, 1, 0) + c_2(0, 1, 1)$ gives $c_1 = x$, $c_2 = z$, $y = x + z$. The span is the plane $x - y + z = 0$.

4. For $q = \alpha + \beta x + \gamma x^2$, solve $a + b(1 + x) + c(1 + x + x^2) = q$: $c = \gamma$, $b = \beta - \gamma$, $a = \alpha - \beta$. A solution always exists.

5. No. The zero matrix is not invertible. (Also $I + (-I) = 0$ shows that addition is not closed.)

6. ($\Leftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, a subspace. ($\Rightarrow$) Suppose neither inclusion holds: choose $\mathbf{u} \in W_1 \setminus W_2$ and $\mathbf{w} \in W_2 \setminus W_1$. If $\mathbf{u} + \mathbf{w} \in W_1$, then $\mathbf{w} = (\mathbf{u} + \mathbf{w}) - \mathbf{u} \in W_1$, a contradiction; if $\mathbf{u} + \mathbf{w} \in W_2$, then $\mathbf{u} = (\mathbf{u} + \mathbf{w}) - \mathbf{w} \in W_2$, a contradiction. So $\mathbf{u} + \mathbf{w} \notin W_1 \cup W_2$, and the union is not closed under addition.